

Virgo Chat Training Summary

Notebook

virgo-chat-training(2).ipynb

Phase

Supervised Fine-Tuning (Virgo Chat)

Purpose

Fine-tunes Virgo Base into Virgo Chat using the prepared conversational dataset while preserving the pretrained language capabilities.

Objective

Load the pretrained Virgo Base model, train on chat conversations, evaluate performance, and save checkpoints for continued instruction tuning.

Input Resources

Pretrained Virgo Base model (virgo_base_final.pt), train.bin, val.bin and virgo_tokenizer.json.

Model Initialization

Loads Virgo Base weights into the same decoder-only Transformer architecture with weight verification before training.

Training Dataset

Memory-mapped binary datasets (train.bin and val.bin) loaded using PyTorch DataLoaders for efficient GPU training.

Dataset Split

Training Dataset: 99% | Validation Dataset: 1%.

Training Configuration

AdamW optimizer, cosine learning-rate decay with warmup, gradient accumulation, mixed-precision training (FP16/BFloat16), automatic mixed precision (autocast) and gradient scaling.

Evaluation

Validation computes cross-entropy loss and token-level accuracy. Includes text generation using temperature sampling and Top-k decoding for qualitative evaluation.

Checkpointing

Saves latest checkpoint, best-performing model, optimizer state, scheduler state and scaler state. Supports automatic training resume.

Outputs

virgo_chat_last.pt, virgo_chat_best.pt, training_state.pt and generated sample responses.

Engineering Highlights

Pretrained weight loading, parameter grouping for weight decay, LambdaLR scheduler, checkpoint recovery, GPU optimization, mixed precision, sequence generation and scalable fine-tuning pipeline.

Summary

This notebook converts Virgo Base into Virgo Chat through supervised fine-tuning on conversational data while maintaining efficient training, evaluation and checkpoint management.